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INVITED TALK / ABSTRACT

Genus two embedded minimal surfaces in $\mathbb{S}^3$ with bi-dihedral symmetry

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Abstract

The isometry group of the classical Lawson embedded minimal surface $\xi_{2,1} \subset \mathbb{S}^3$ of genus 2 has order 48 and it is isomorphic to the product $D_4 \times S_3$ of the dihedral group D_4 of order 8 and the permutation group S_3 of order 6. $D_4 \times S_3$ has three subgroups of index 3, all of which are conjugate to one another and all isomorphic to the bi-dihedral group $D_{4h} = D_4 \times \mathbb{Z}_2$ of order 16. We will discuss a recent work with José Espinar, on how to characterize $\xi_{2,1}$ among closed embedded minimal surfaces of genus 2 in $\mathbb{S}^3$ whose isometry group contains D_{4h}.

Keywords: genus two embedded minimal surfaces; $\mathbb{S}^3$; bi-dihedral symmetry; Lawson embedded minimal surface; isometry group; dihedral group; permutation group; bi-dihedral group